DocuMind

Enterprise Document Intelligence System using RAG and LLMs

Abstract

The increasing volume of unstructured data, such as PDFs and documents, creates challenges in extracting meaningful insights efficiently. This project presents DocuMind, an intelligent document question-answering system built using a Retrieval-Augmented Generation (RAG) framework. DocuMind enables users to upload documents and ask natural language questions, returning accurate, context-aware responses grounded in the document content. By combining semantic retrieval using vector databases with Large Language Models (LLMs), the system reduces hallucination and improves reliability. The solution is implemented using LangChain, FAISS, HuggingFace embeddings, LLaMA (Ollama), and Streamlit, demonstrating a practical end-to-end AI application.

1. Introduction

Organizations generate large volumes of unstructured documents, making it difficult to locate relevant information quickly. Traditional keyword-based search systems lack semantic understanding and often fail to capture context.

To address this challenge, DocuMind leverages advancements in Generative AI and Retrieval-Augmented Generation (RAG) to build a system that understands document content and answers user queries intelligently.

Unlike standard LLM-based systems, DocuMind ensures that responses are grounded in retrieved document content, thereby enhancing trust and accuracy.

2. Objectives

The key objectives of this project are:

- Develop a system capable of understanding and processing document content
- Implement semantic retrieval using vector embeddings
- Generate accurate answers using LLMs based on retrieved context
- Reduce hallucination through context grounding
- Provide a user-friendly interface for real-time interaction

3. System Architecture

DocuMind follows a modular **RAG-based architecture** consisting of three main components:

1. Document Processing Layer

- PDF upload and parsing
- Text extraction using document loaders
- Chunking into smaller segments with overlap for context preservation

2. Retrieval Layer

- Conversion of text chunks into embeddings
- Storage in FAISS vector database
- Retrieval of top-k relevant chunks based on similarity

3. Generation Layer

- Prompt construction using retrieved context
- Answer generation using LLaMA (via Ollama)
- Display of answers along with source chunks for transparency

4. Methodology

4.1 Document Processing

Uploaded PDFs are parsed and split into smaller overlapping chunks to maintain contextual continuity and improve retrieval precision.

4.2 Embedding Generation

Each chunk is transformed into a vector representation using a pre-trained embedding model. This enables semantic comparison rather than keyword matching.

4.3 Semantic Retrieval

User queries are converted into embeddings, and the most relevant chunks are retrieved from FAISS using similarity search.

4.4 Answer Generation

A structured prompt combines the retrieved context and the user query. The LLM generates responses strictly based on this context.

4.5 Prompt Engineering

Multiple prompt strategies were explored:

- Basic prompting
- Chain-of-thought prompting
- Few-shot prompting

These strategies improve clarity, reasoning, and response structure.

5. Results and Observations

The system successfully demonstrated:

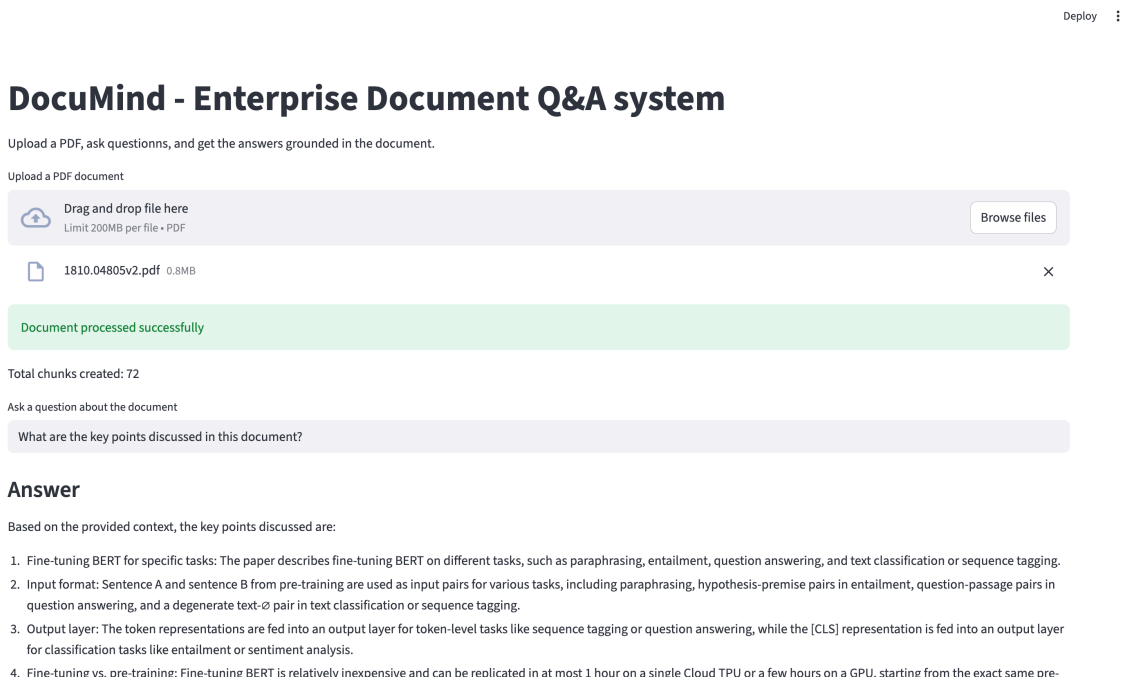
- Accurate retrieval of relevant document segments
- Context-aware answer generation
- Reduced hallucination through grounding
- Improved response clarity using prompt engineering

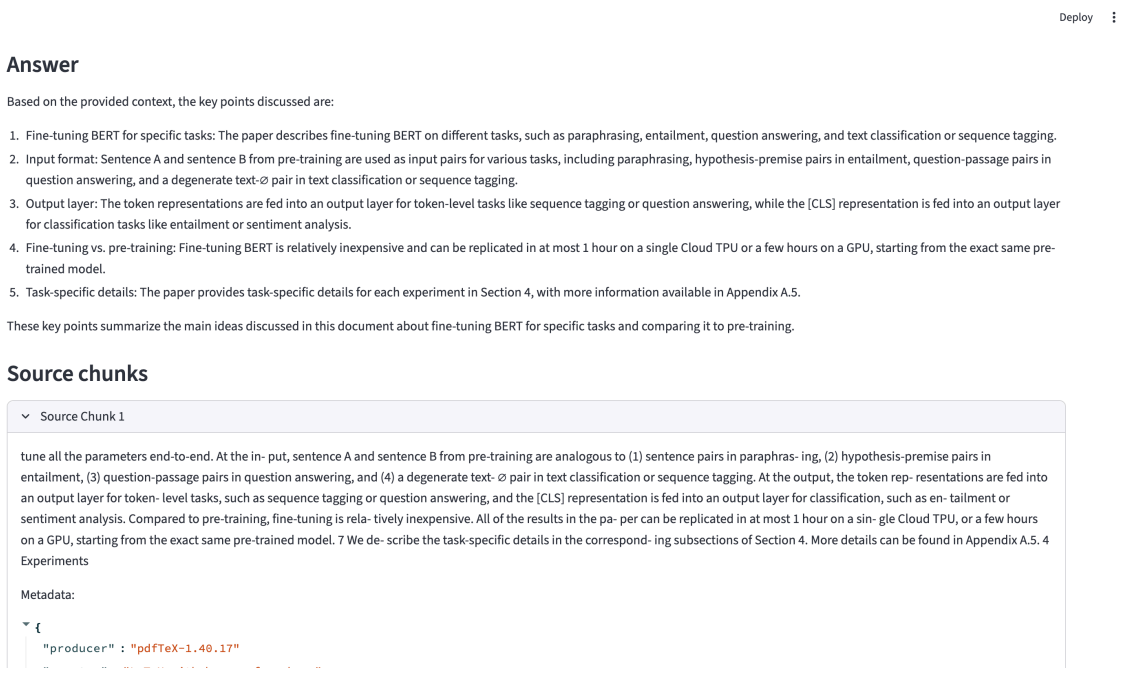
The inclusion of **source chunks alongside answers** enhanced transparency and user trust, allowing verification of generated outputs.

7. Future Work

- Integration with cloud platforms such as Azure
- Support for multiple document formats (DOCX, TXT)
- Automated evaluation metrics (e.g., faithfulness, relevance)
- Chat-based interface with memory

- Hybrid retrieval (vector + keyword search)





7. Conclusion

DocuMind demonstrates the effectiveness of combining retrieval-based systems with large language models for document understanding tasks. By grounding responses in retrieved context, the system enhances reliability and reduces hallucination, making it suitable for real-world applications such as enterprise knowledge systems, document search engines, and AI assistants.